B.Tech Operating Systems End-Semester Exam 2024

1. (a) Explain the safety algorithm and resource-request algorithm in Banker's Algorithm. [8]
(b) Solve: 4 processes and 3 resources with given Allocation and Max matrices. Is the request (1,0,2) by P1 granted? [10]
2. Explain the deadlock detection algorithm for multiple instances of resource types. [8]
3. Draw the Gantt chart and calculate average waiting time for 5 processes using Round Robin with quantum 2 ms. [10]
4. Explain paging hardware with TLB. [6]
5. Explain the Readers-Writers problem. [6]
6. Explain linked and indexed file allocation. [5]